Subdivision Surfaces

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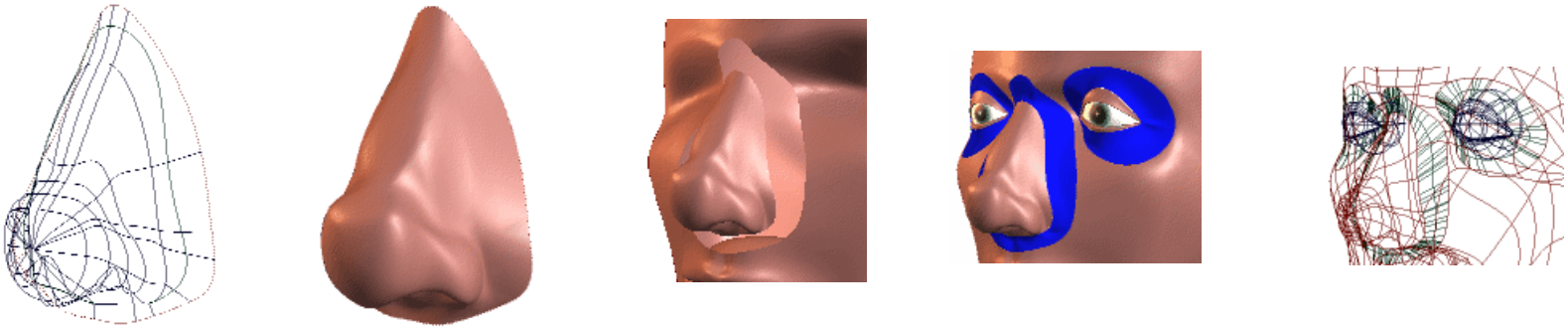
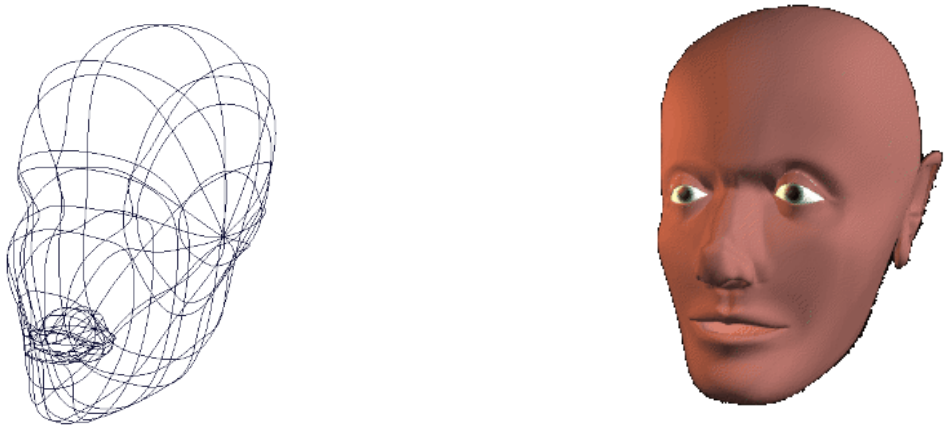
Motivation

- Vast literature on surface modelling via polynomial or rational patches
- High quality and precision
- Rigid shape (rectangular or triangular)
- Geometric constraints between neighbours to achieve smoothness

Motivation





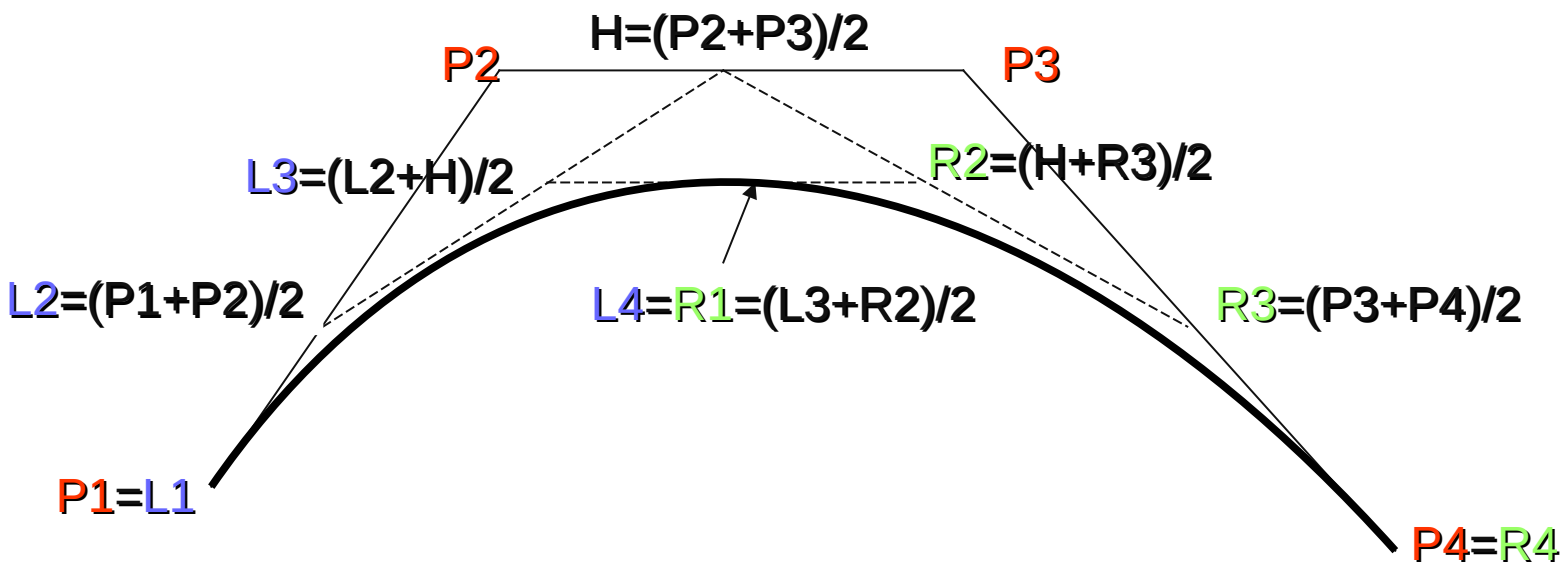
Goal

Represent a curved surface

- Smoothness
- Affine invariance
- Compact representation
- Efficient rendering
- Easy editing (intuitive shape control)

Curves

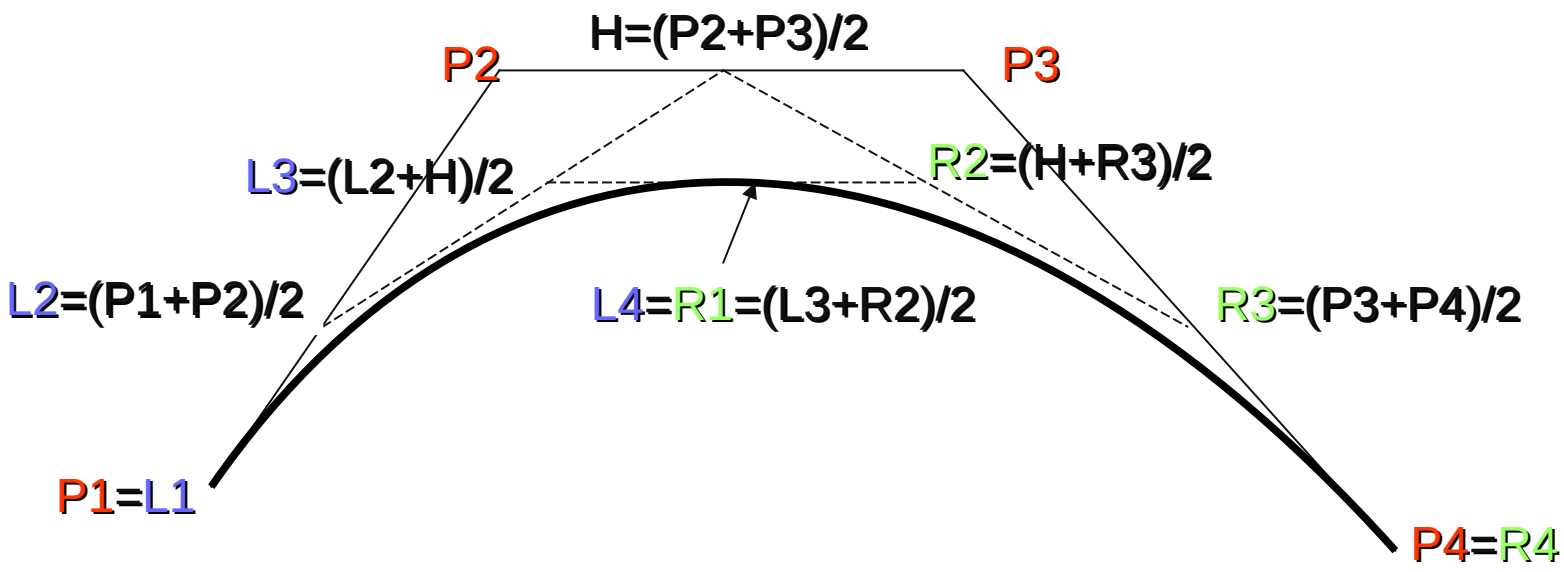
- De Casteljau subdivision



example for t = 0.5

Curves

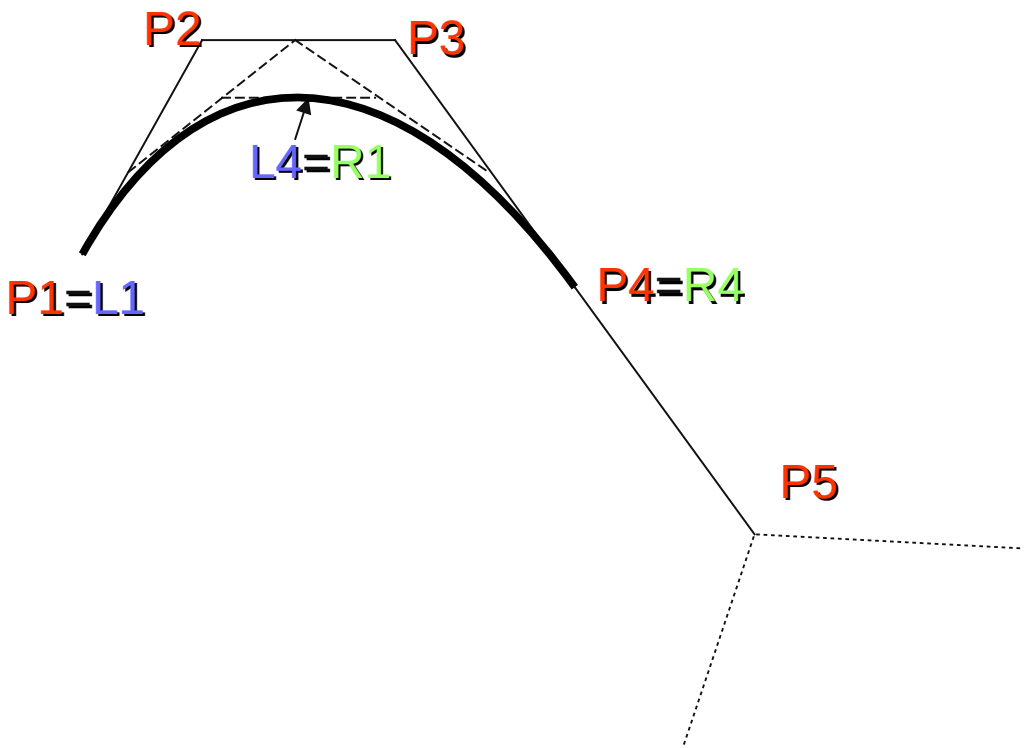
- De Casteljau subdivision



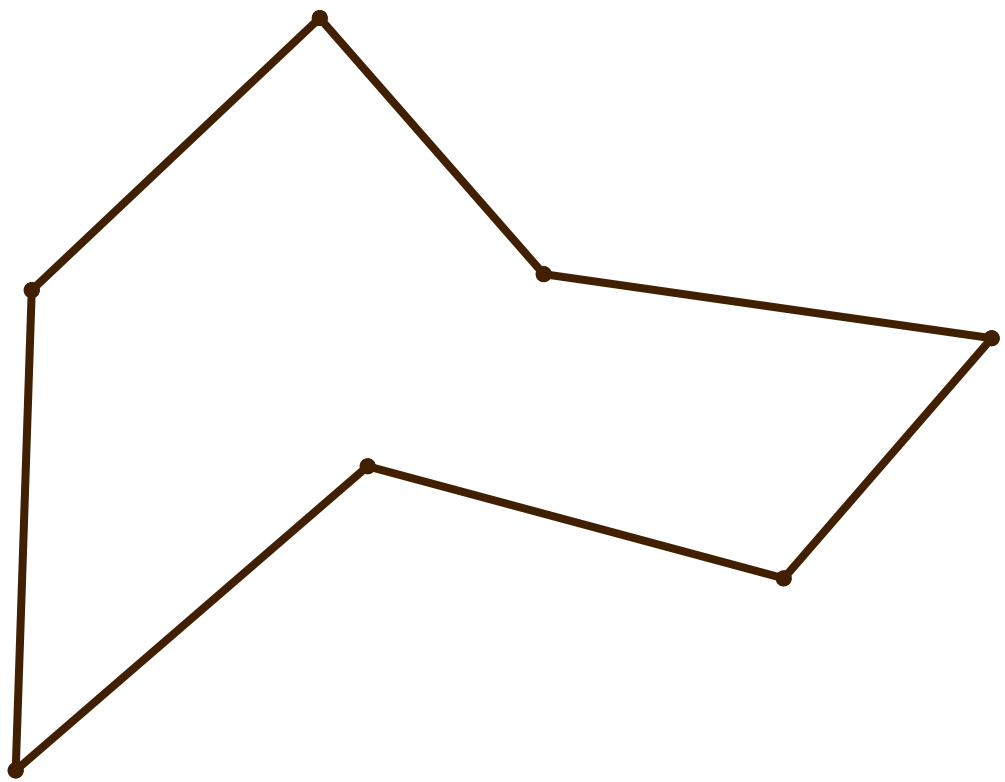
example for t = 0.5

Curves

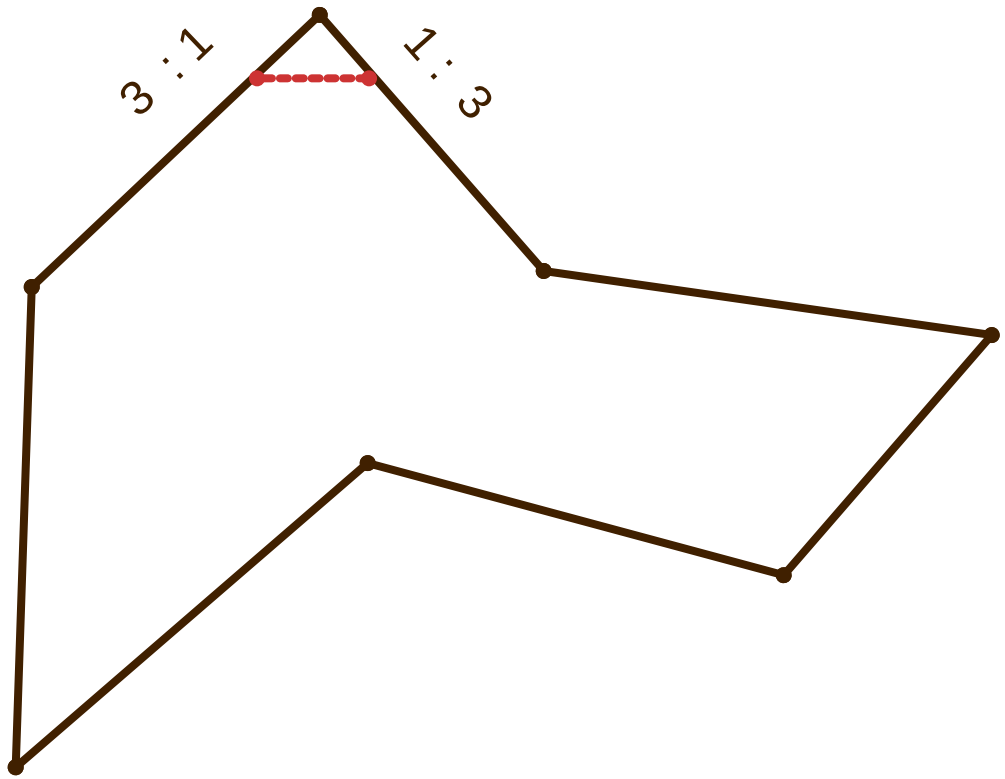
- Next point is determined by C^1



Corner Cutting [Chaikin’74]



Corner Cutting



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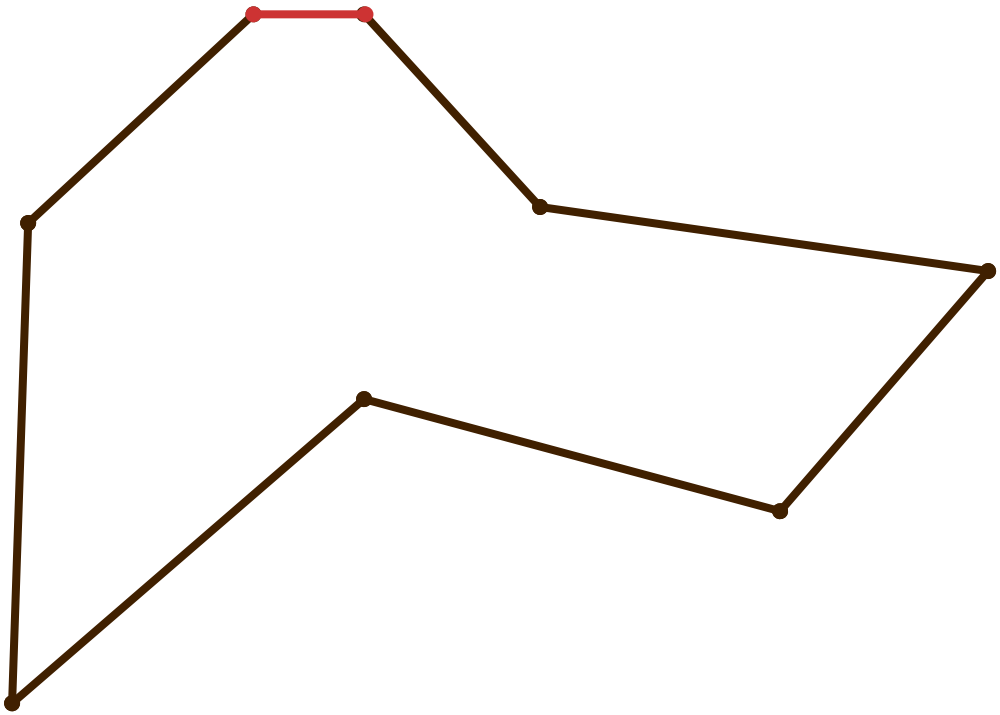

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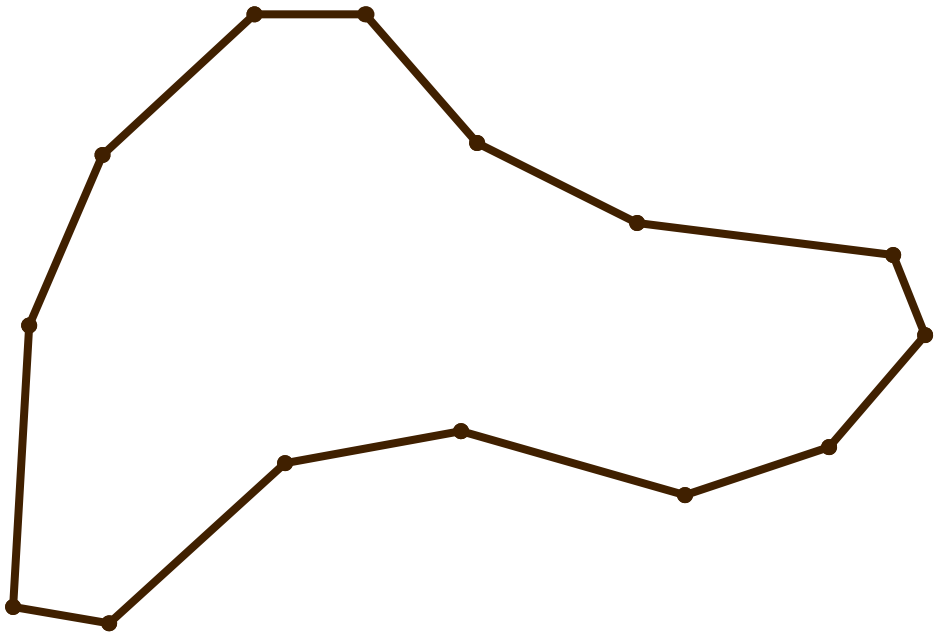

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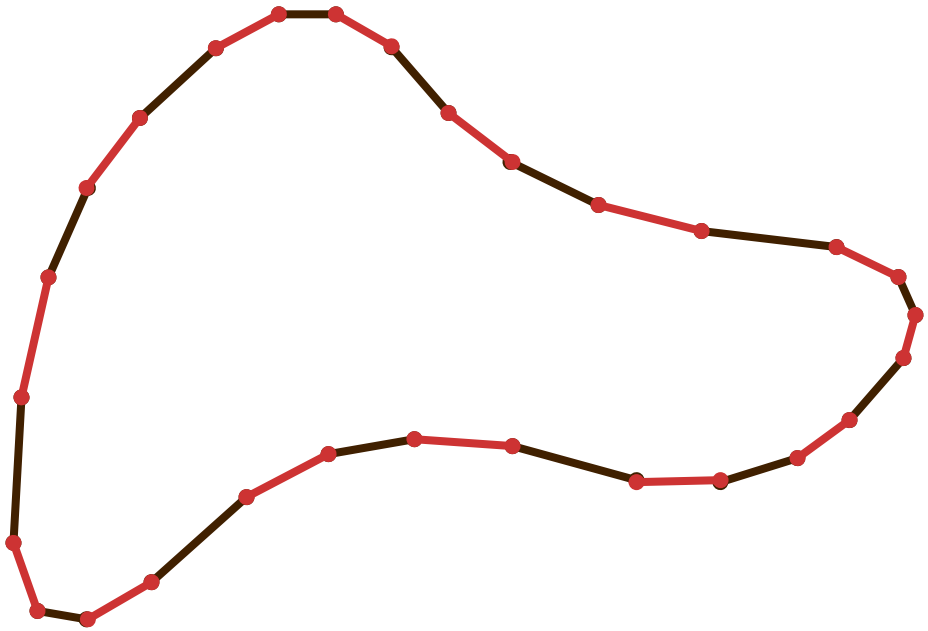

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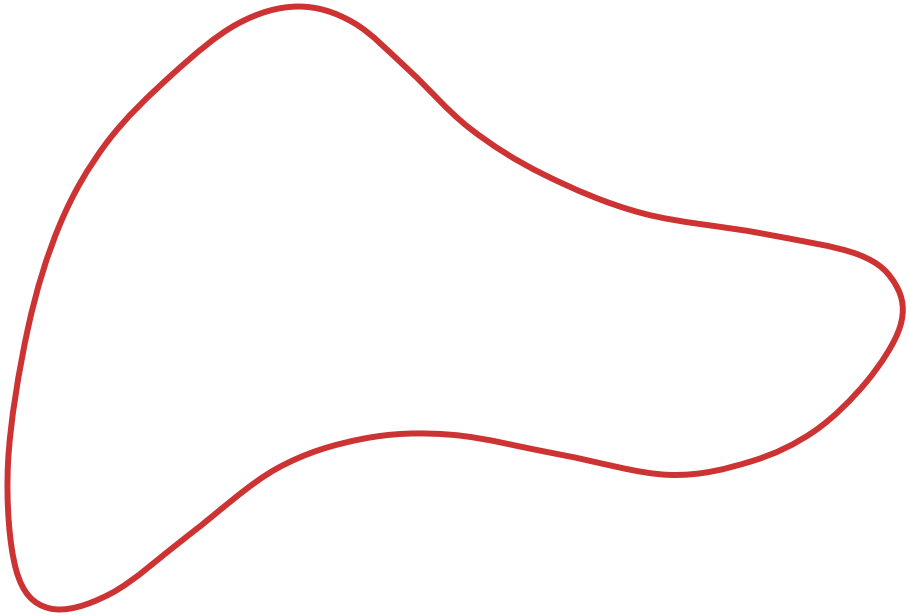

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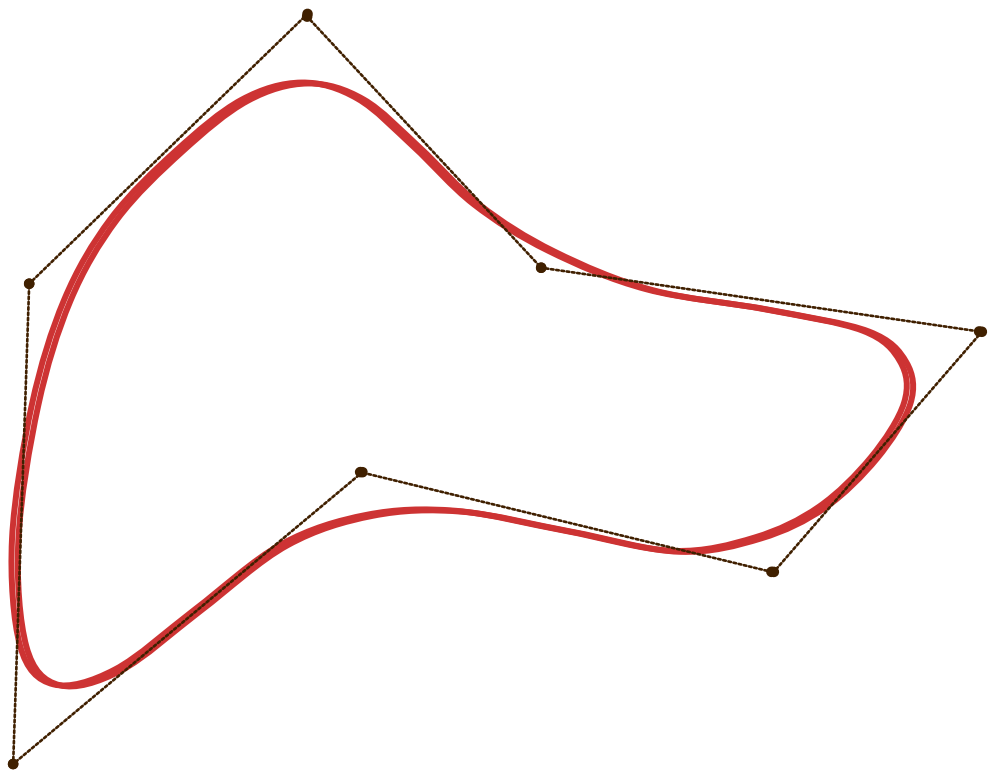
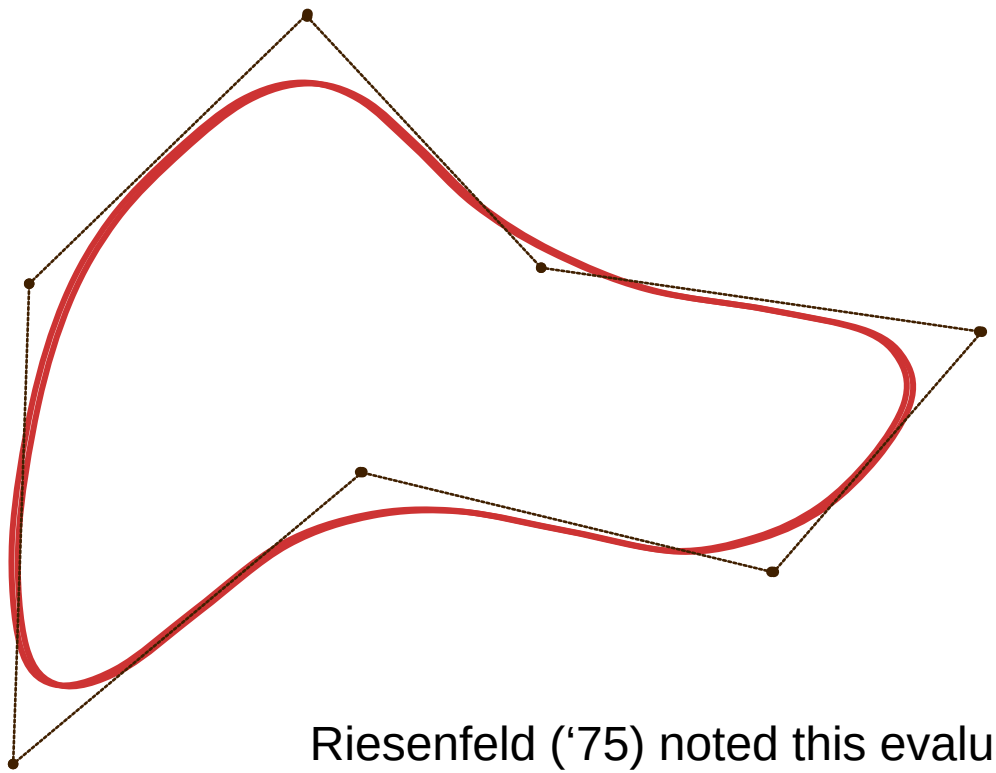
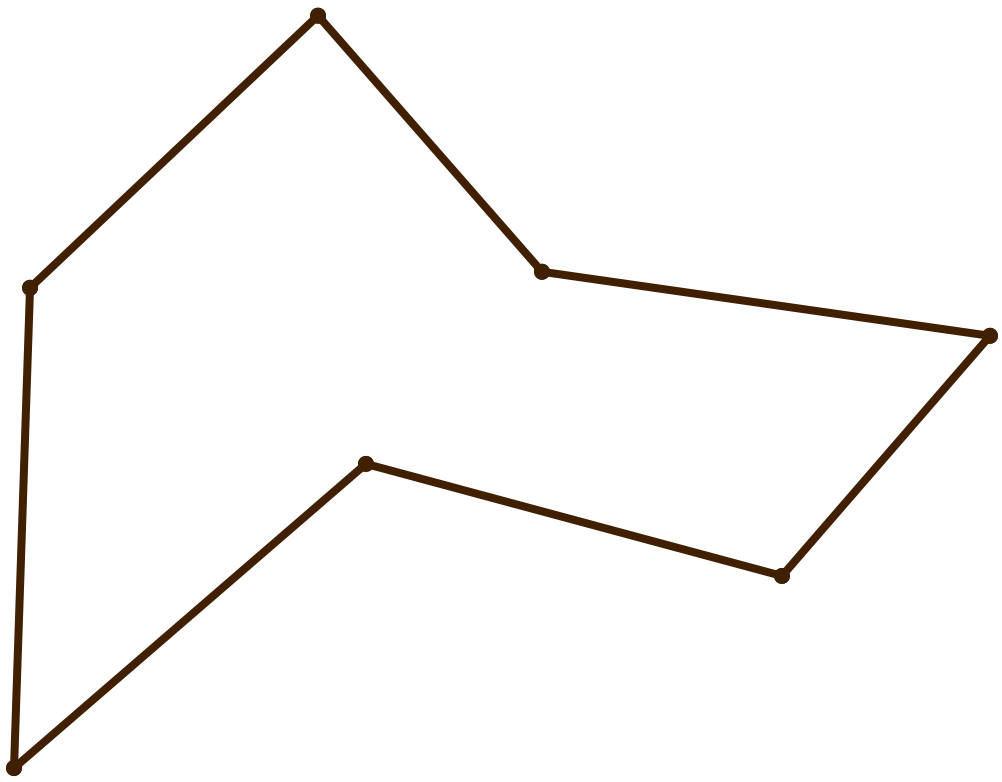


Corner Cutting

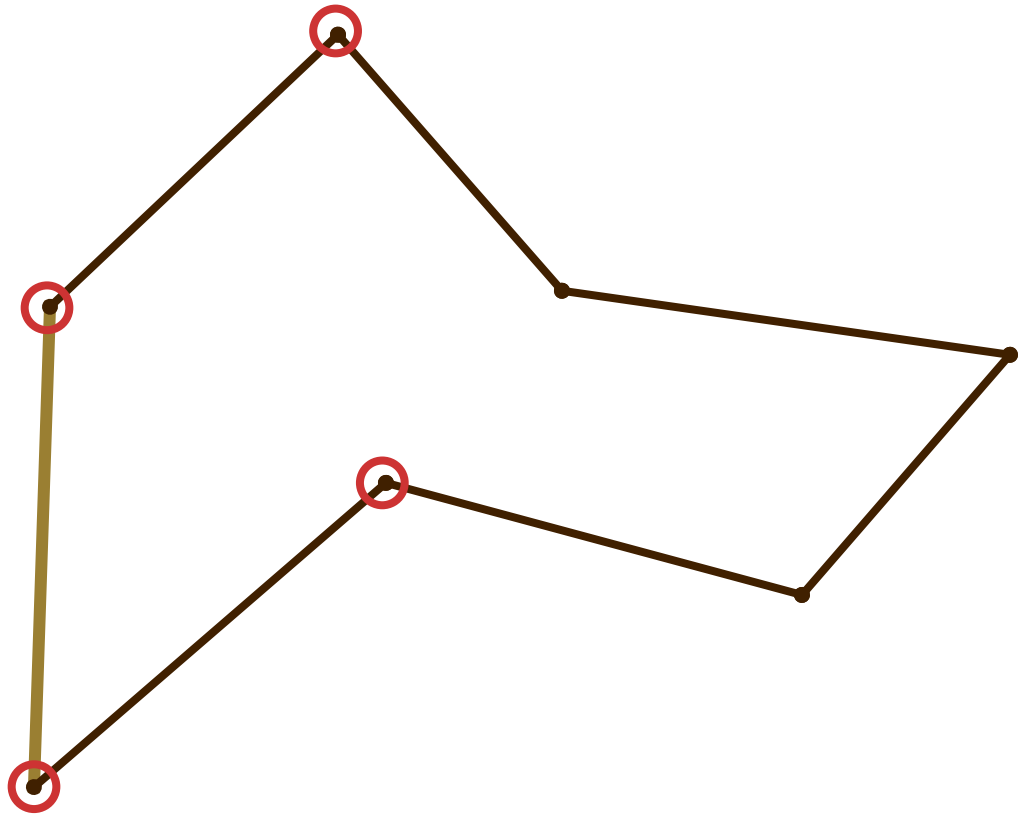


Riesenfeld (‘75) noted this evaluates
Quadratic B-splines

4-point scheme



4-point scheme



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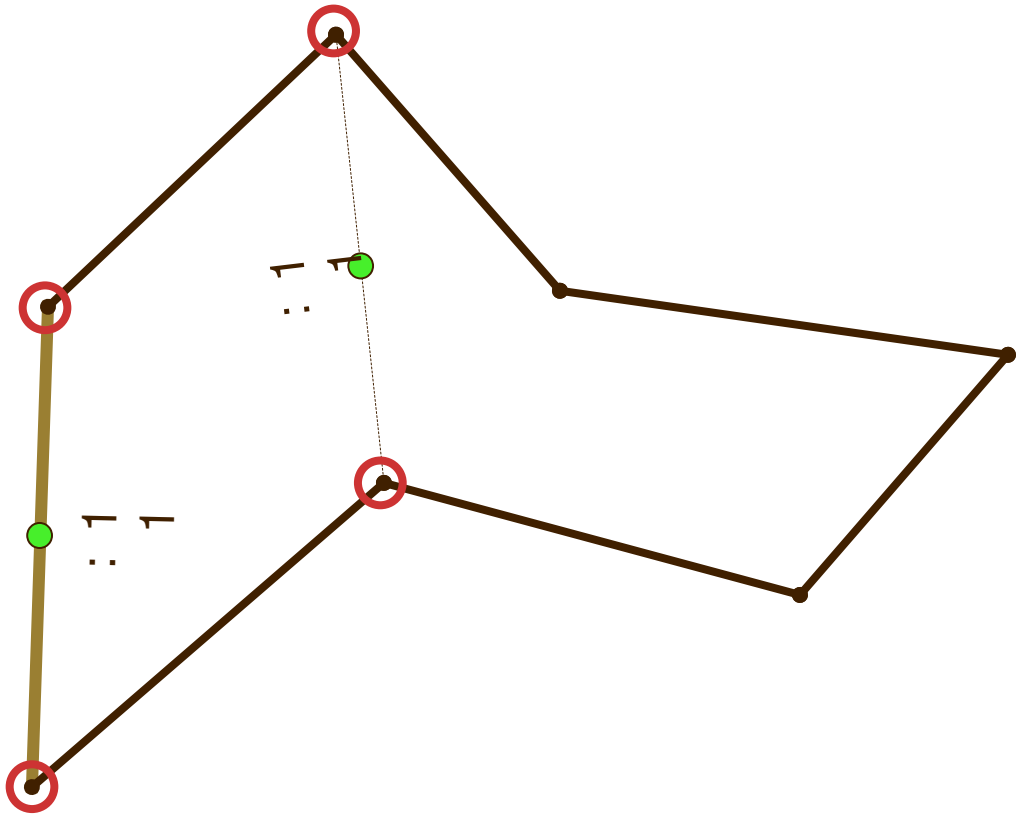
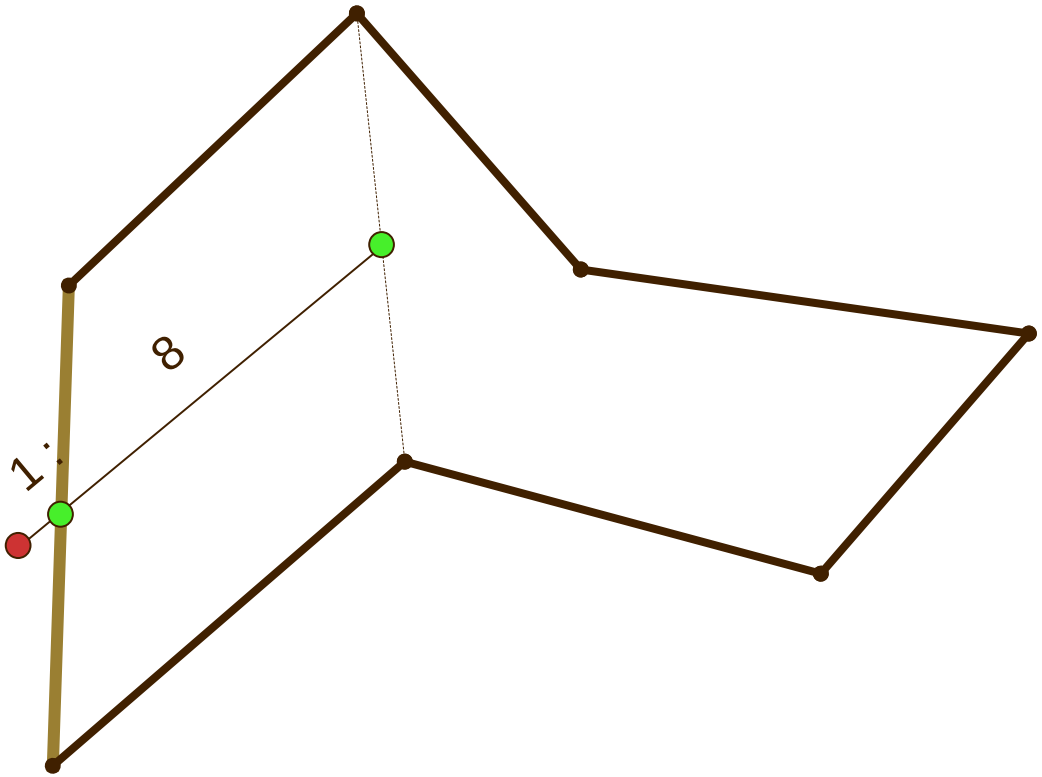


4-point scheme



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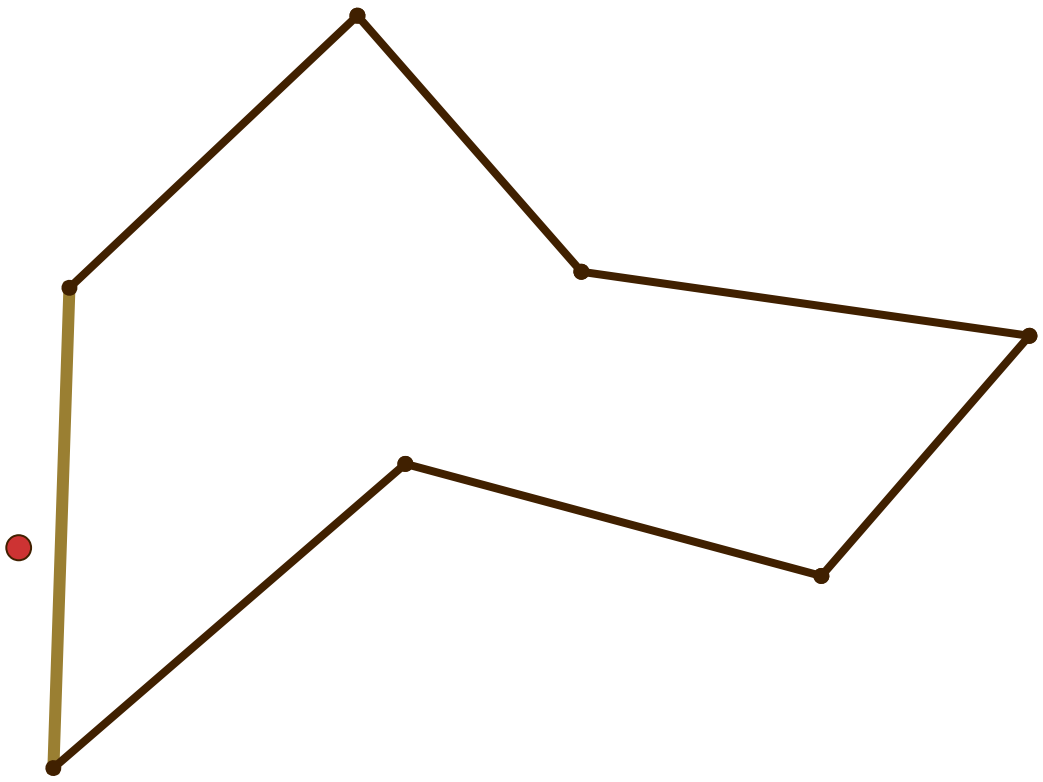

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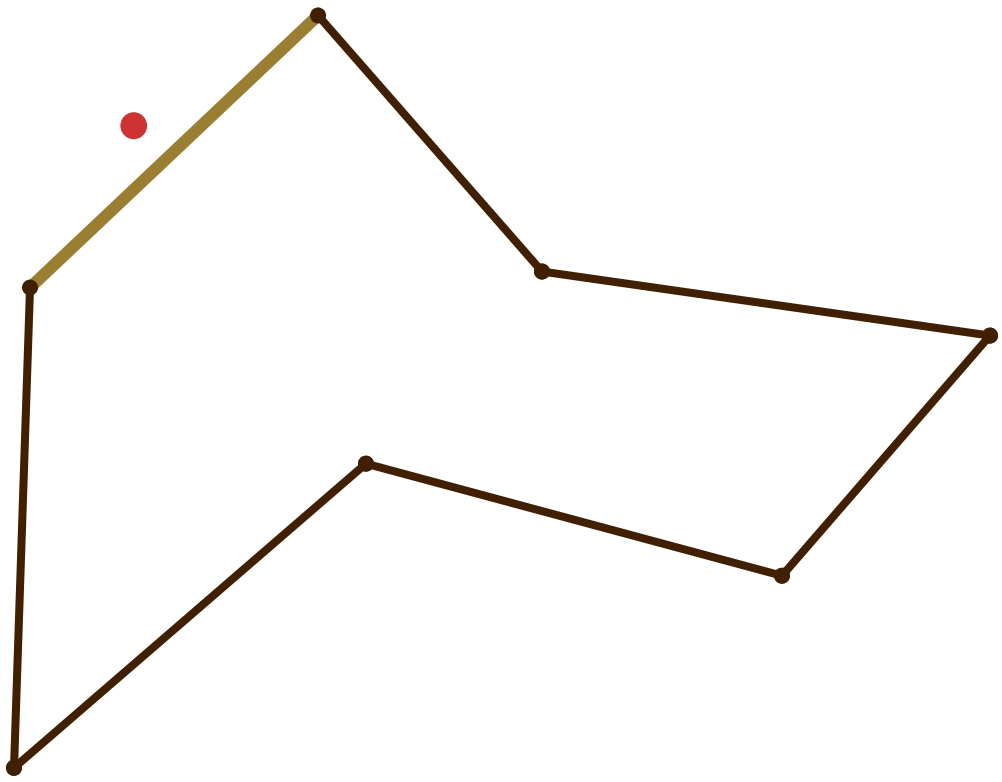

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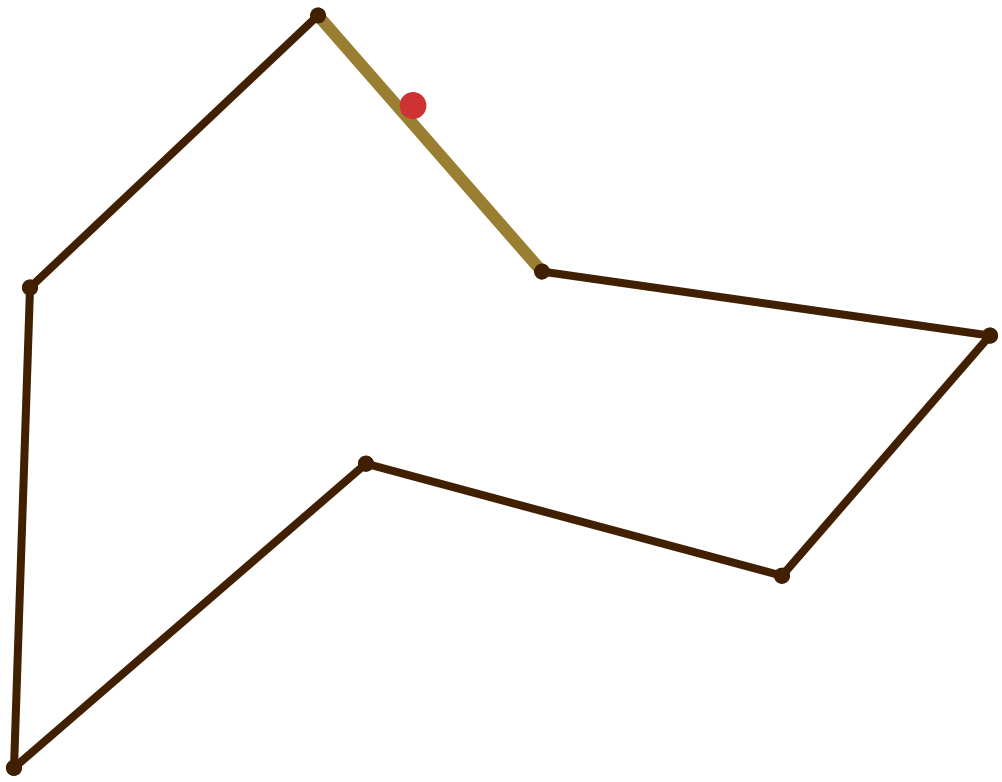

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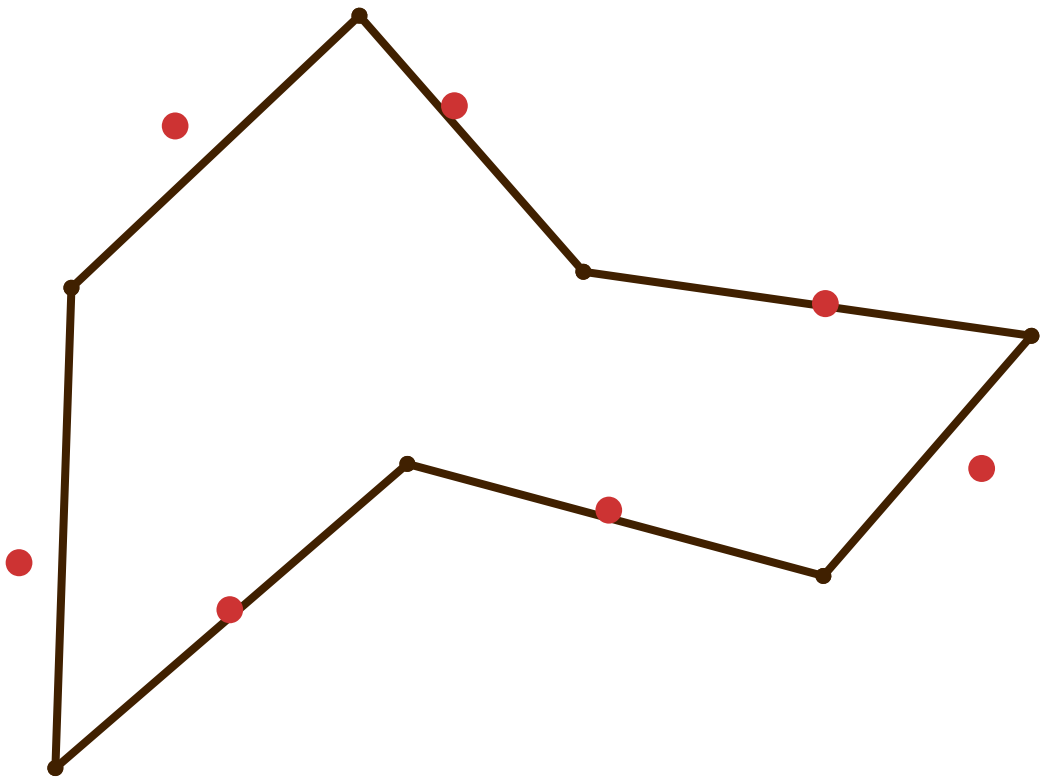

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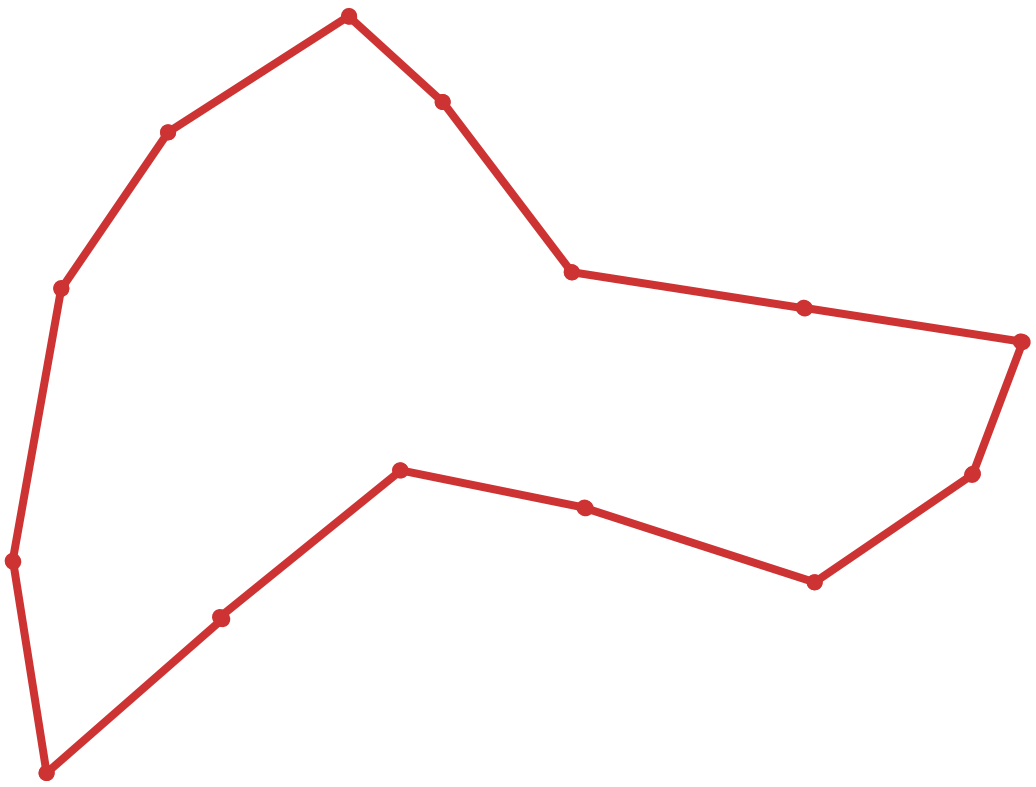

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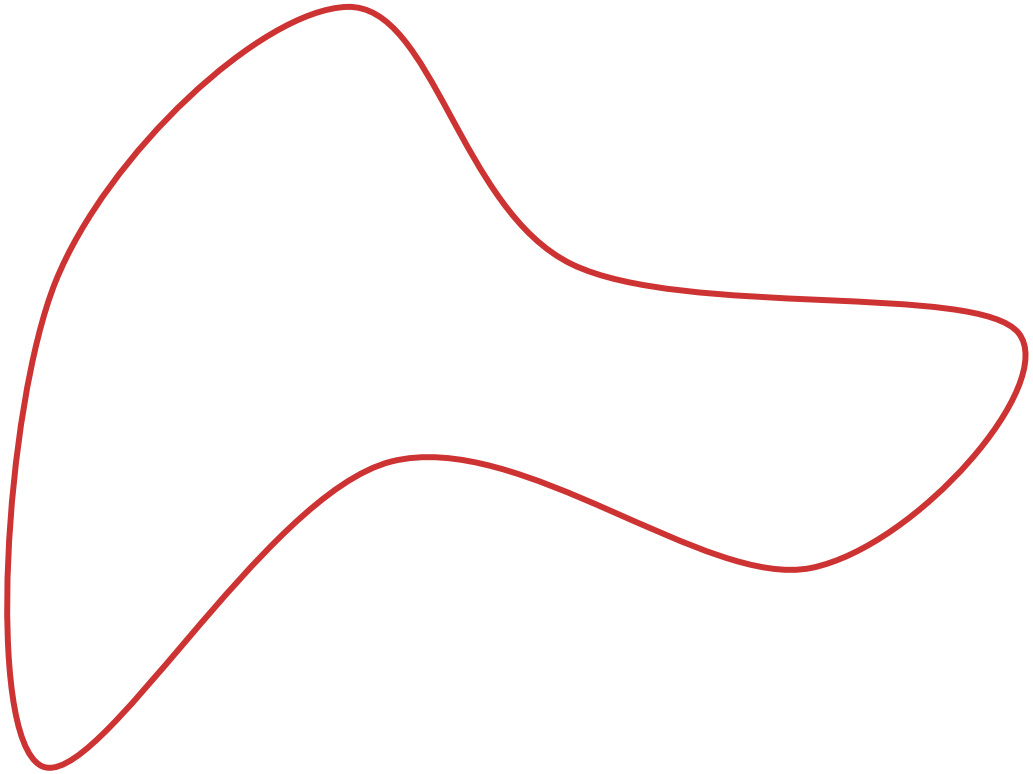

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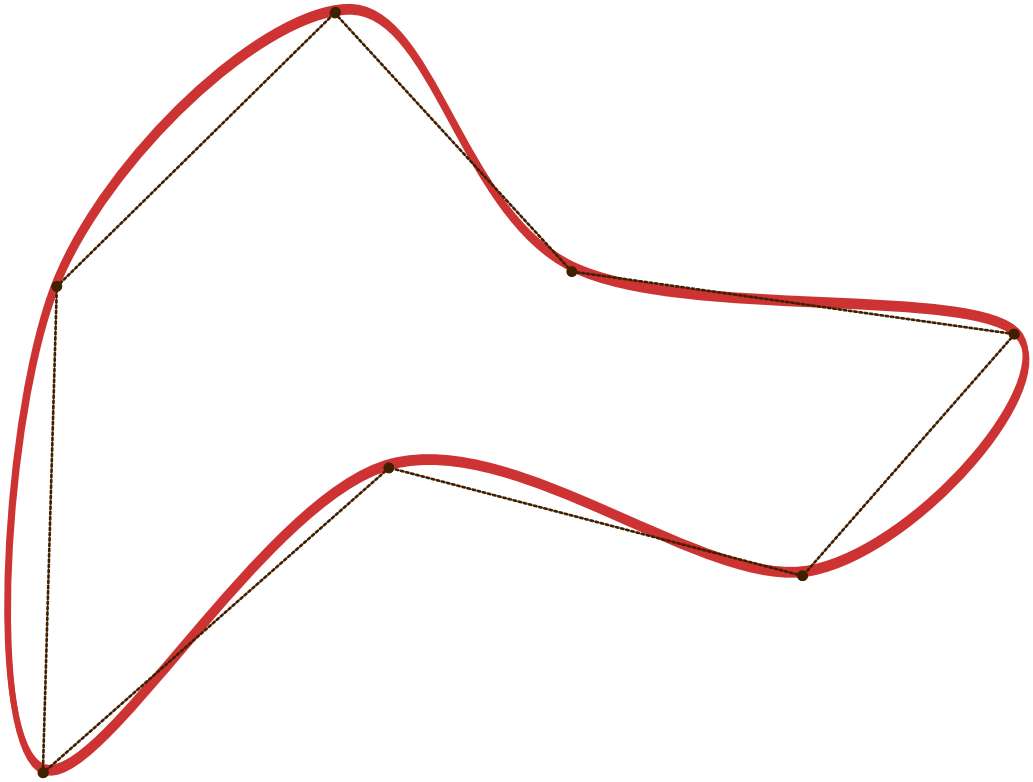
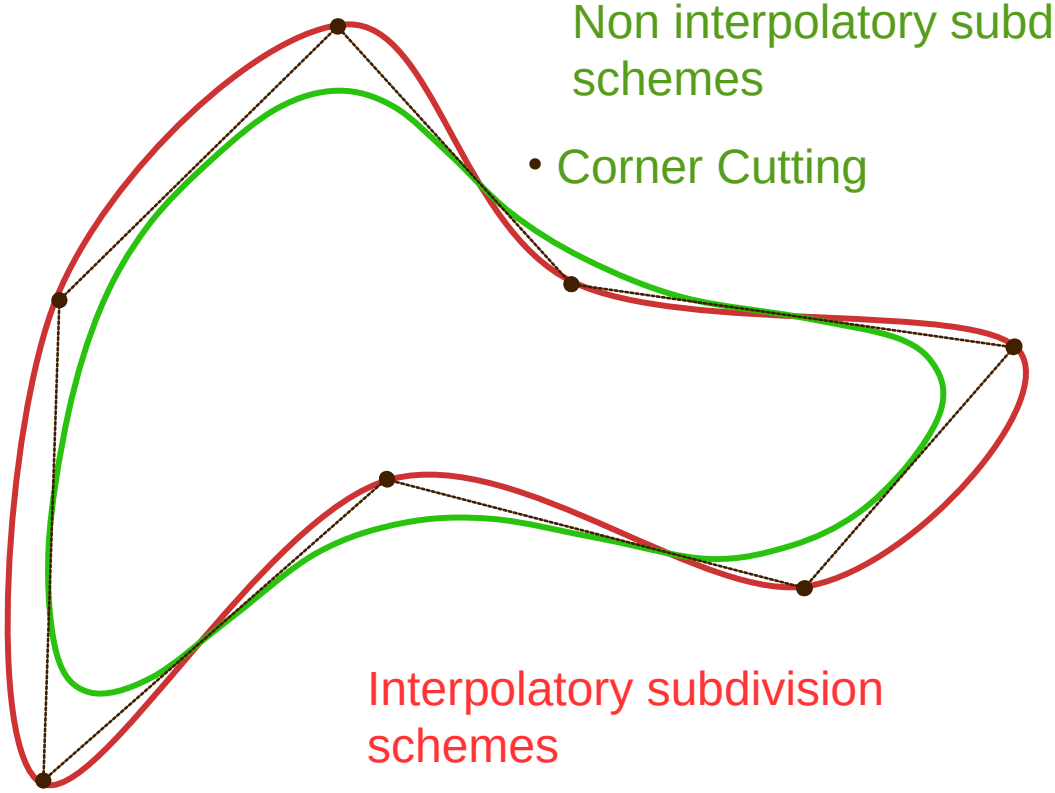


Curve subdivision



Non interpolatory subdivision schemes

- Corner Cutting

Interpolatory subdivision schemes

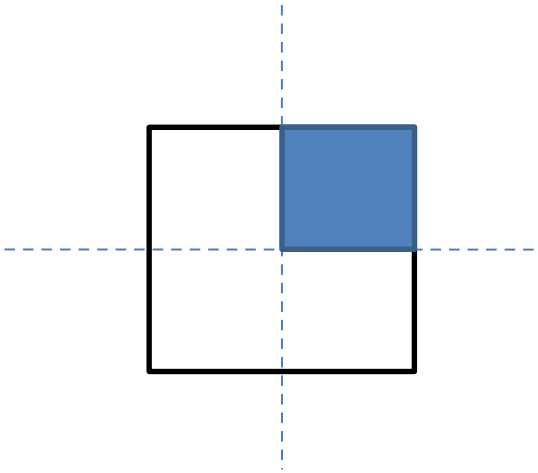
- The 4-point scheme

Surfaces

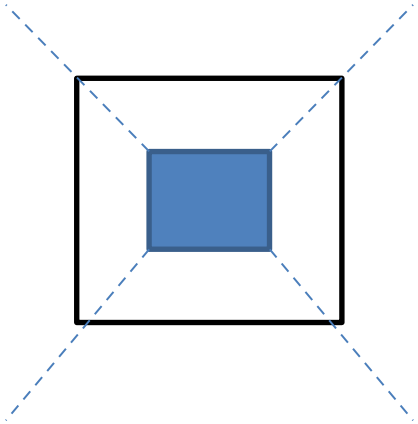
Geri’s game

A bug’s life

Two strategies

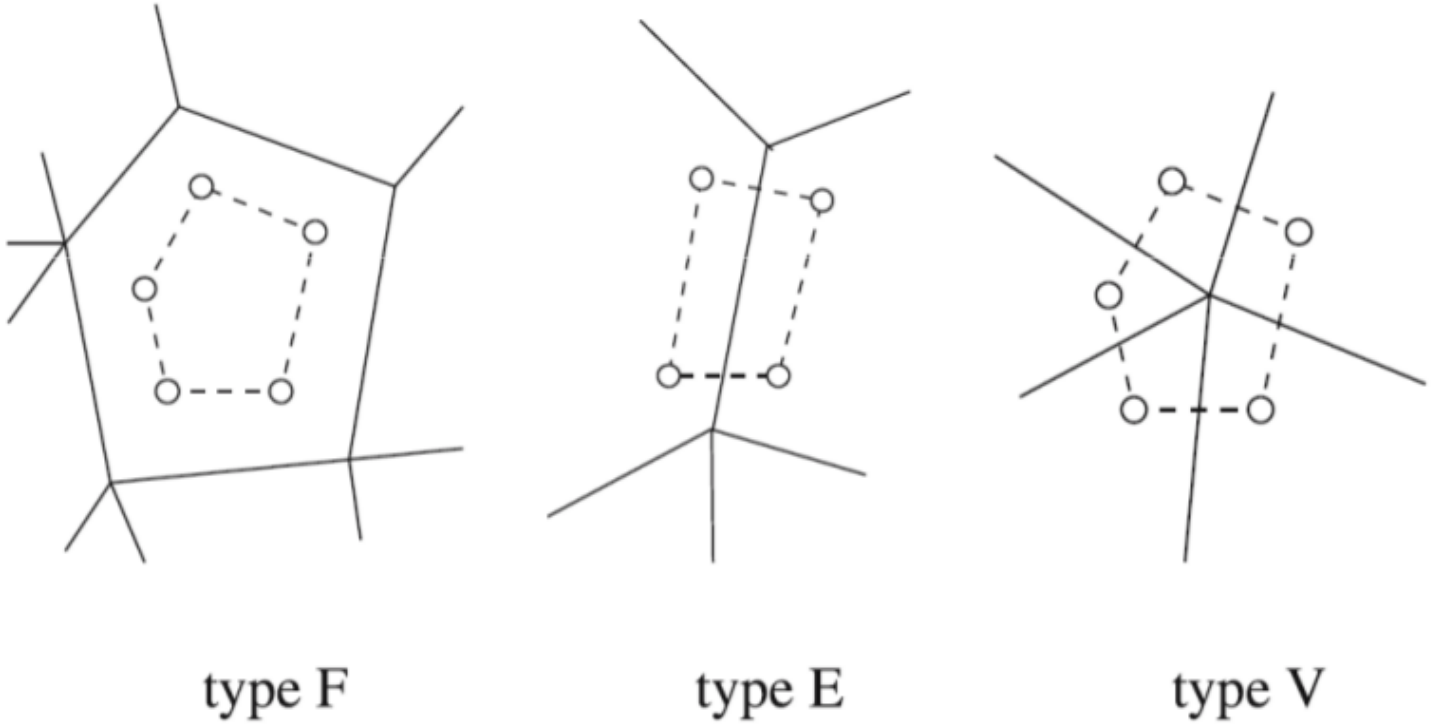


face split



vertex split

Doo-Sabin: vertex-split, any polygon



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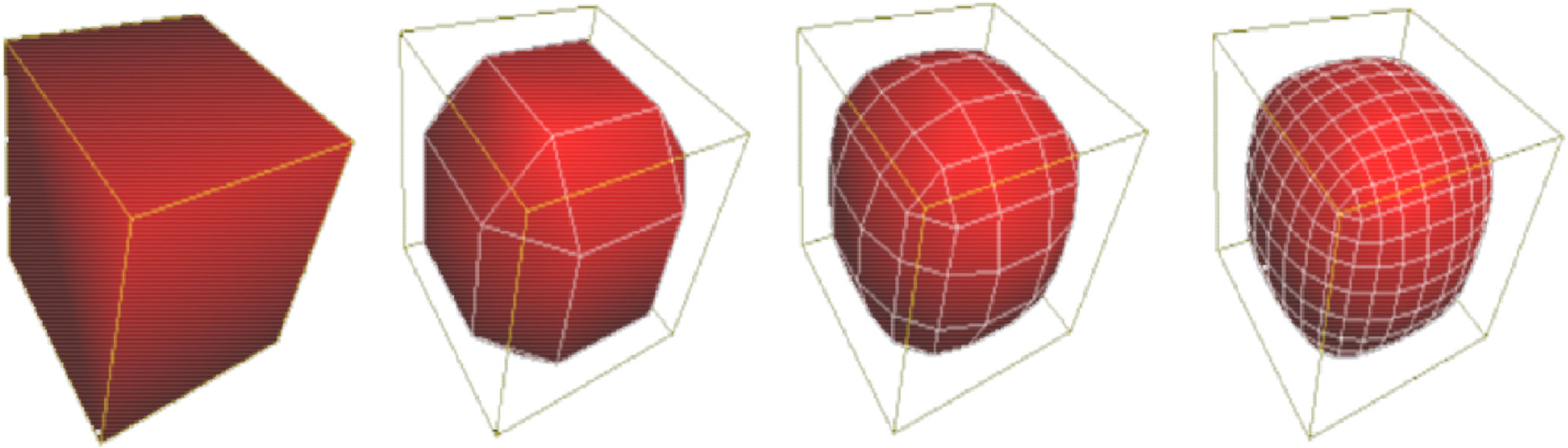

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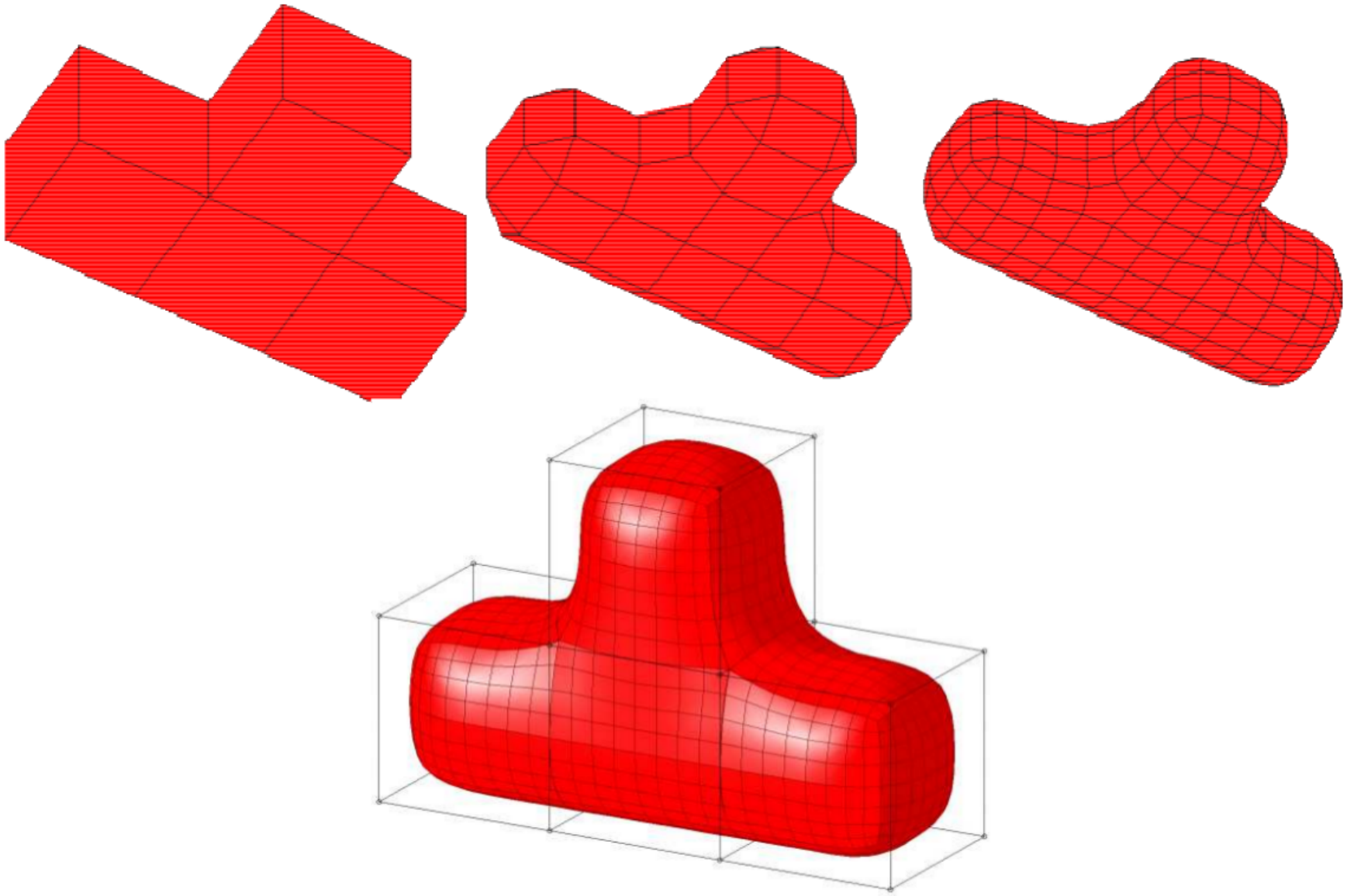
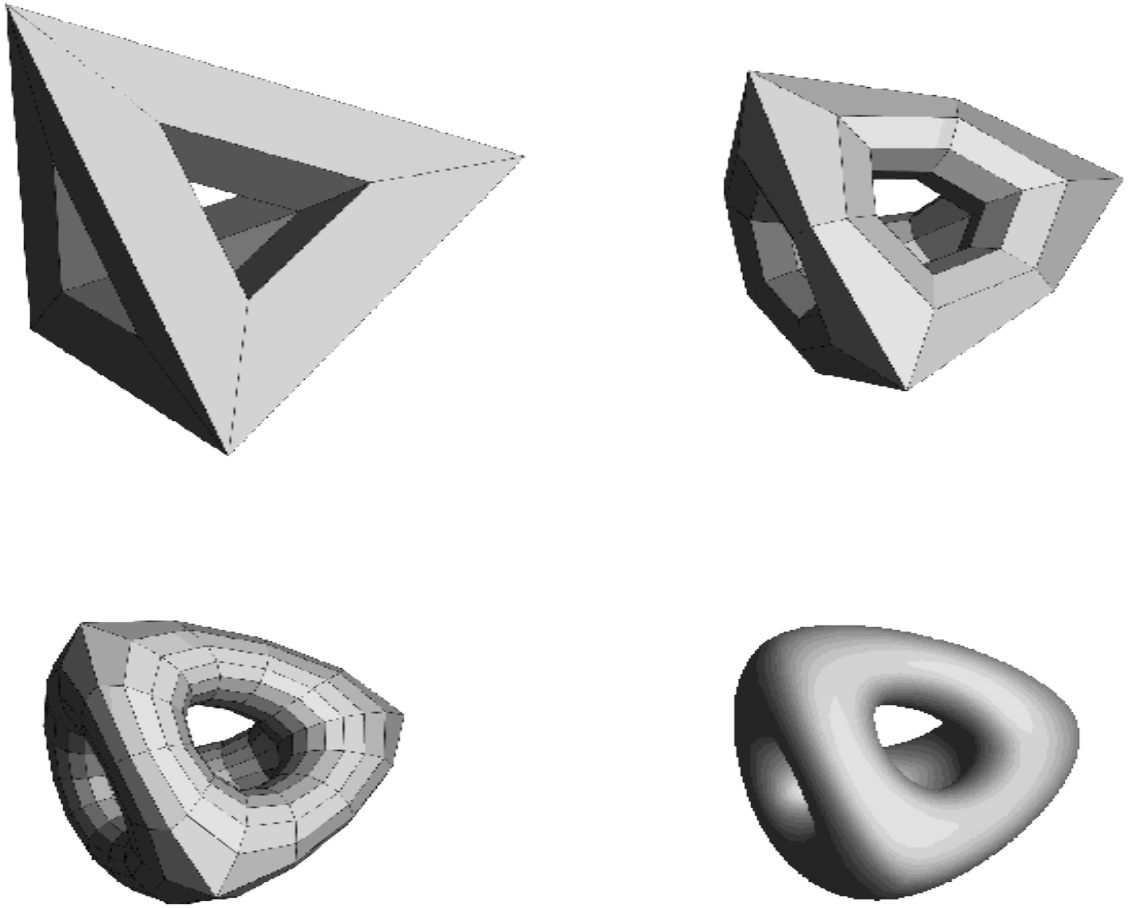


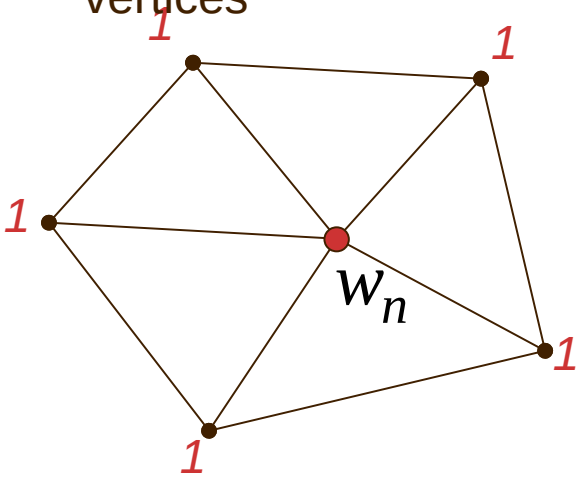
Catmull-Clark: approx. face-split



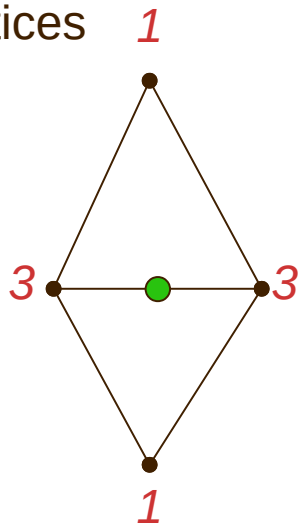
Loop: approx. face-split for tri. meshes

Every new vertex is a weighted average of the old vertices.

A rule for new red vertices



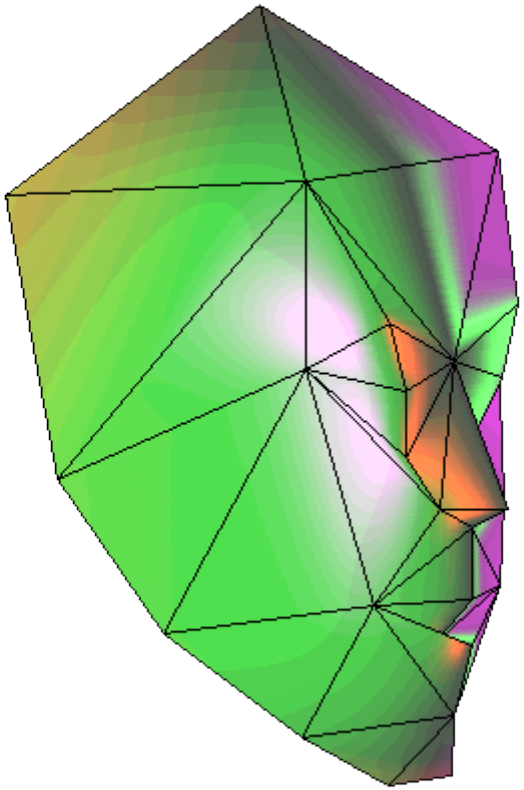
A rule for new green vertices

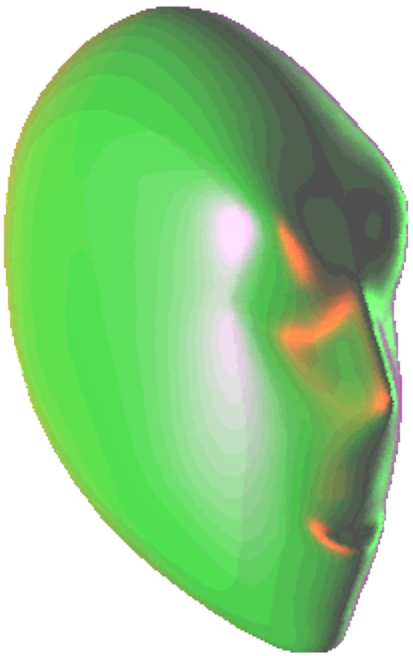


n - the vertex valency

$$w_n = \frac{64n}{40 - (3 + 2\cos(2\pi/n))^2} - n$$

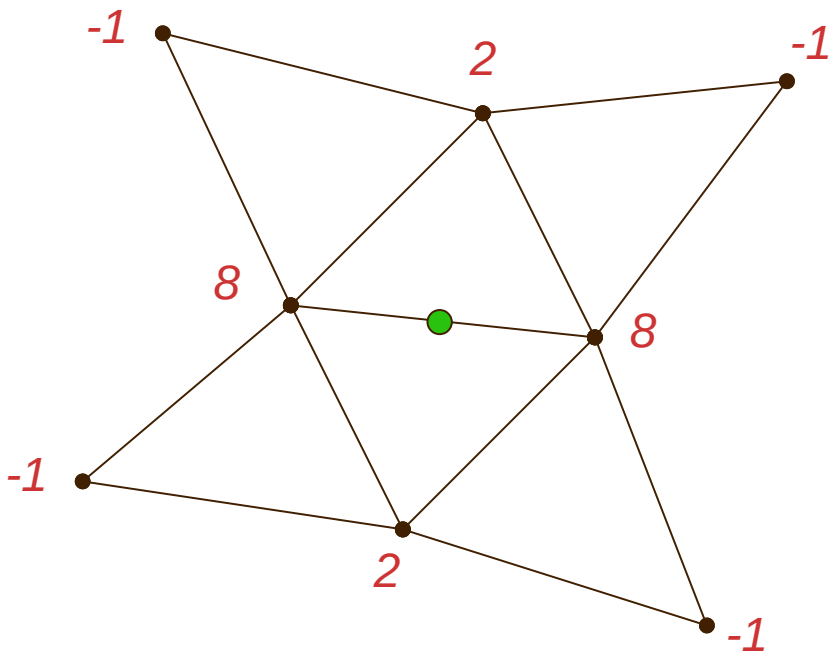
Loop subdivision



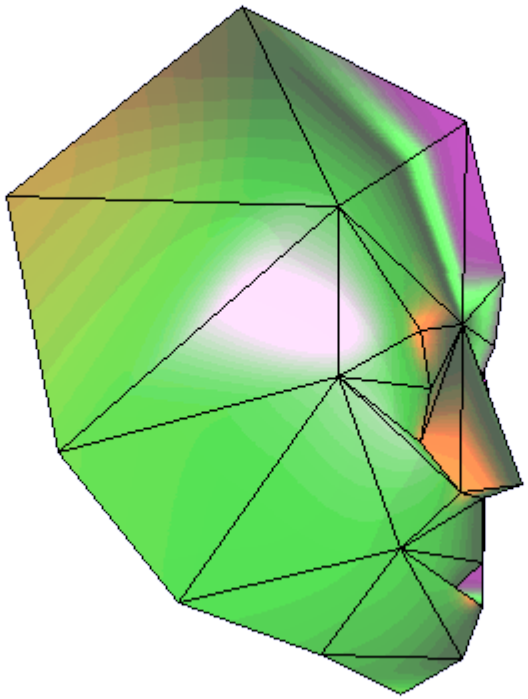


Limit surface has continuous curvature almost everywhere

Butterfly: Interpolating face-split for tri. meshes



Butterfly



Only VC^1 continuity

Catmull-Clark: details

Geometry changes:

1. Add F "**face" vertices**
2. Add E **"edge" vertices**
3. Move the V original vertices (**"vertex" vertices**)

Geometry changes

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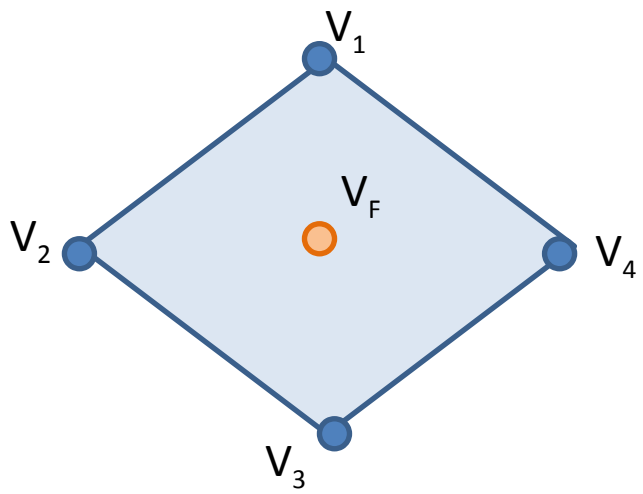
V_F = centroid of face F

$V_F = (V_1 + V_2 + \ldots + V_n)/n$



Geometry changes

1. Add F "**face**" **vertices**
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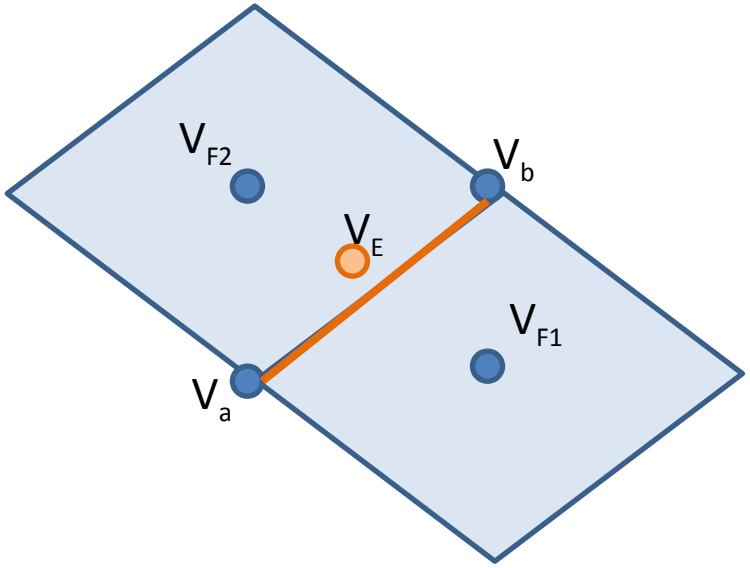
$V_E = (V_a+V_b+V_{F1}+V_{F2})/4$



Geometry changes

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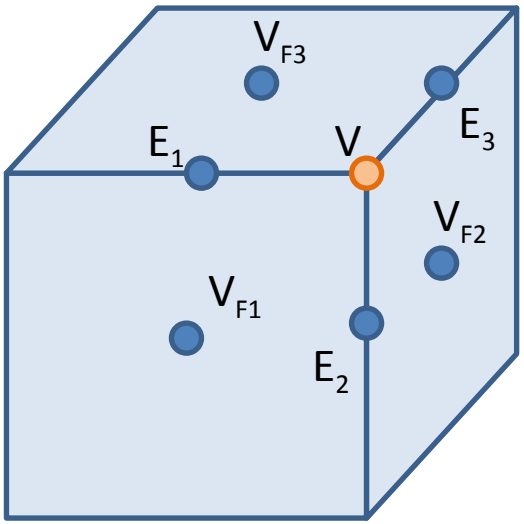
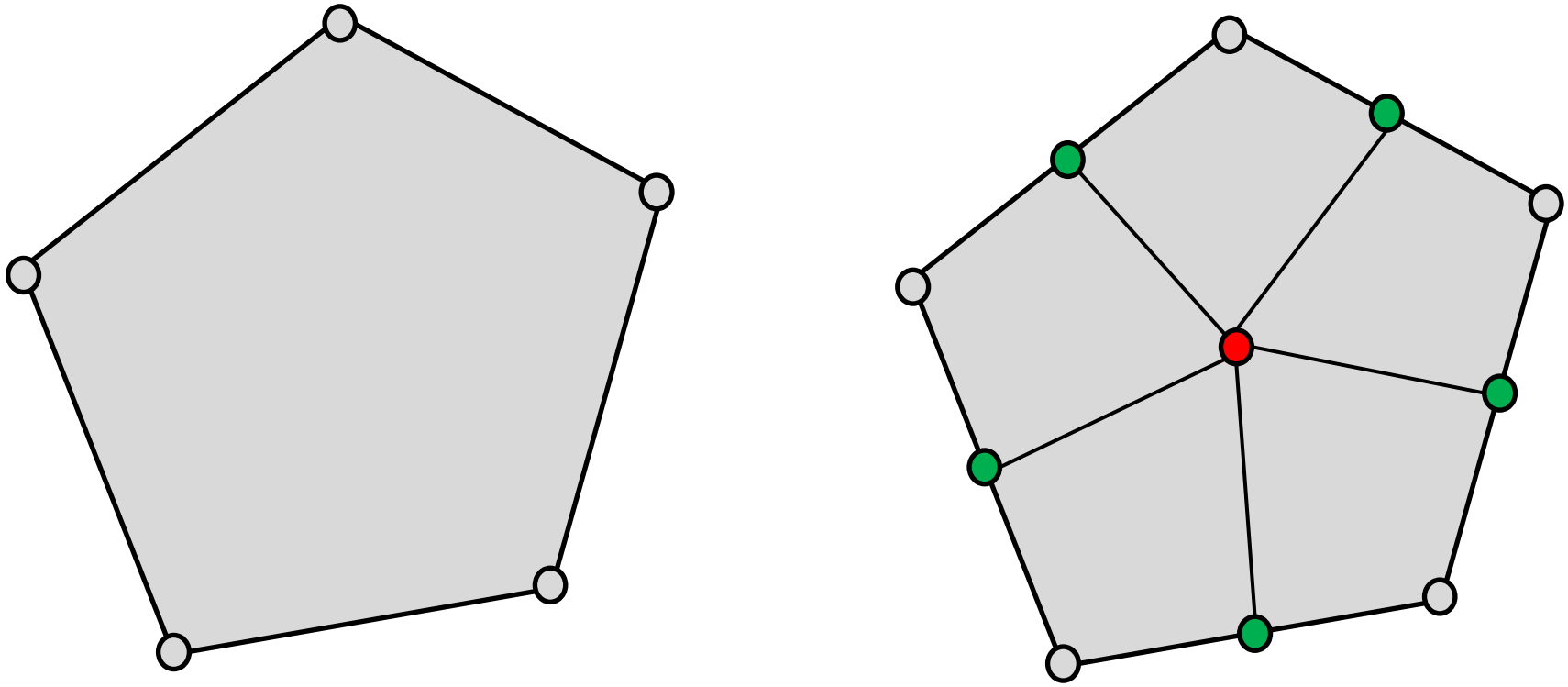
$V_V = (F+2R+(m-3)V) / m$

F=centroid of V_F vertices

R= centroid of edge midpoints

V= original vertex

m=degree of vertex V



Topology changes